ZOO501 CURRENT MCQS MIDTERM (2018) UPDATED

1. Red blood cells make Globins
2. how many strategies for developmental commitment 2
3. Ribosomes produce in sea urchin 10^{12}
4. Resact is isolated from egg jelly of sea urchin
5. ___is a discipline that studies embryonic & other developmental processes Developmental Biology
6. 0nly small %age of genome expressed in particular cell
7. Digestive & respiratory system formed by Endoderm
8. Progesterone work for signals the egg to resume its meiotic division
9. Insect characteristics are ..... specification syncytial
10. In Chlamydomonas haploid chromosome has.....copy one
11. Modes of commitments are..... 3
12. After fertilization developmental process proceed by Cleavage
13. Ca ion responsible for cortical granules stores in Endoplasmic Reticulum of Eggs
14. Seawater has a particularly high ___ ion concentration Na (Sodium)
15. There are ___approaches of anatomy 4(four)
16. The major cell adhesion molecules appear to be the Cadherins.
17. It is estimate ____egg is 25000 to 50000 mRNA. *Sea Urchin*
18. Fusion of ____and sperm cell plasma membrane. Egg
19. Resact diffuse readily in sea water.
20. The requirement of capacitance varies from species to Species.
21. The division of embryo into 3 Germ layers
22. Cell potency ___≥___fate
23. Zona pellucida also initiate the acrosomal reaction after sperm bound to it.
24. Mammalian egg is surrounded by layer called. Cumulus
25. The volume of the sea urchin is about 200 Picoliters.
26. The sperm travel by the amoeboid motion of lamellipodial extensions of the cell membrane.
27. The sex and Reproduction process included into fertilization.
28. The term spermatozoa means.. Sperm Animals
29. In 1824 J. L. Prevost and J. B. Dumasclaimed sperm is not parasite.

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30. 4 cadherin classes are found in Vertebrates embryos. .
31. The second major group of protostomes are the Lophotrochozoa
32. Sperm discovered in 1678
33. Transcription factors have three major domains
34. How many Amphibian oocytes produces ribosomes as 10^{12}
35. Enhancers can also be used to inhibit transcription
36. In deutrosomes the mouth opening would formed after... opening Anal
37. Germs cell formed in.... specialized portion of egg cytoplasm
38. Syncytial specification found in insects
39. Third anatomical approach to developmental biology is Teratology
40. The mature sea urchin is metabolically sluggish and activated by. Sperm
41. The resting membrane potential is.....mv. 70mV
42. Fusion is a Active process that is activated by fusogenic proteins.
43. Axoneme s formed by. Microtubules
44. Acrosome can be modified secreted by..... Golgi apparatus (not sure)
45. The male leopard frog release sperm in the summer.
46. Eukaryotic genes are contained within a complex of DNA and protein called Chromatin
47. The negative enhancers are called Silencers
48. The ability of embryonic cells to change their fates is called..... Regulation
49. Pax6 transcription factor found in mammalian eye, nervous system and pancreas.
50. E- Cadherin is expressed on all early mammalian embryonic cells.
51. Annelids, molluscs and flatworms belong to..... Lophotrochozoa
52. Potency of cell decrease if cell is committed.
53. Fertilization in Rana Pipiens usually lasts for about 3 (three) years.
54. An uncommitted cell can be described as Naive
55. Lying immediately beneath the plasma membrane of the egg is a thin Shell
56. Acroomal vesicles are considered as modified secretory vesicles
57. Most marine organisms release their gametes into the environment
58. The fast block method is used to prevent Polyspermy
59. Autonomous specification was first demonstrated in 1887
60. In mouse ZP3 is attached to zona pellucid
61. TAFs used in transcription factor activity
62. in slow block method which is cortical granule protein hyalin and peroxidase
63. Egg shell formed weight of egg. 9 – 12%
64. potency is intrinsic property
65. cytoplasm that contains many nuclei called syncytium
66. Ectoderm forms..... epidermis, nervous system, and pigment cells.
67. according to dutch sperm are ... parasitic animals (spermatozoa)
68. Acrosome & nucleus constitute the head of the sperm.
69. cortical granules are membrane bound structures
70. Acrosomal reaction has 2 (two) component
71. Acrosome-reacted sperm transfer their binding from ZP3 to the adjacent ZP2 molecules
72. Ca & bicarbonate ions maybe critical in activating cAMP

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73. Peroxidase enzyme hardens fertilization envelope
74. The Fast block to polyspermy is achieved by changing the electric potential
75. P-cadherin stands for _____Placental Cadherin
76. Most transcription factor can bind to specific DNA sequences
77. MITF protein necessary for production of pigments cell and their pigments.
78. Outside of plasma membrane is the vitelline envelope.
79. Mouse __fertilin___is localized to posterior plasma membrane of the sperm.
80. Finally the fourth cortical granules protein ___hyaline___ forms a coating around egg.
81. The cleavage stages are called.____Blastomere
82. Enhancers are....._DNA Sequences
83. The mitotic spindle and contractile ring are perpendicular to each other
84. J.Whittaker stained blastomeres for the presence of the enzyme acetyl cholinesterase
85. Protocadherins are involved in separating notochord from mesoderm in __Xenopus gastrulation
86. Pax6 factor needed for mammalian eye
87. Which stands comparative embryology is ____First_
88. Cell type specification precedes any large-scale embryonic cell migration
89. Series of extremely rapid divisions in zygote is ___Cleavage
90. In protosome, mouth is formed first than Deuterostomes: forms.
91. Northern blots, in situ hybridization, & the PCR can show which cells are transcribing particular genes.
92. Coding regions of DNA are called ___Exon
93. Na and __K__ Are important in fast block polyspermy.
94. The force for sperm propulsion is provided by __dynein
95. Spallenazani demonstrated that filtered __toad__ semen devoid of sperm would not fertilize egg.
96. Mesoderm generates the____KIDNEY.
97. Sperms are attracted towards egg of their species by CHEMOTAXIS.
98. Mathematical modeling seeks to desire developmental phenomenon in terms of EQUATIONS.
99. A region that is not translated in proteins is called UNTRANSLATED region.
100. Invariable cleavage produce the ___SAME___ lineages in embryo
101. Germ layers are formed as a result of ___GASTRULATION
102. RNA polymerase binds with PROMOTOR in start of transcription
103. The region of globular actin molecule lies between nuclear and acrosomal vesicle in ____SEA URCHIN
104. Which technique is used in situ hybridization ____Northern blots
105. Conditional specification involves interaction with RELATIVE POSITIONS cells
106. Particular proteins or carbohydrate on the sperm surface __LOST__ during capacitation
107. ___LOSS____ of hydrogen ion causes pH to rise.
108. Peroxidase enzyme hardens the fertilization envelope by crosslinking TYROSINE residues
109. FERTILIZATION transmits genes from parents to offspring.
110. Regulation of gene expression accomplished by SELECTIVE MESSENGER RNA TRANSLATION
111. Development contact genotype and PHENOTYPE
112. Nucleosome is composed of OCTAMER of histone protein.
113. Knowledge of gene activity in humans can be obtained by candidate GENE MAPPING & POSITIONAL CLONING
114. Anatomical abnormalities are caused by MUTANT GENES
115. All multicellular organisms arise by a slow process of progressive change called DEVELOPMNET

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116. Animals development is the division of embryo into three layers Ectoderm, Mesoderm and ENDODERM
117. The ions which are responsible for increasing granule reactions Ca (Calcium)
118. The zona pellucida is a fglycoprotein matrix, which is synthesized and secreted by the growing oocyte
119. In several other species, certain regions of the membrane are specialized for sperm recognition & fusion
120. An intrinsic program has been activated within the cell that causes it to follow a particular pathway of development
121. Activation of egg metabolism to start development
122. Insects such as Drosophila use all three modes of specification to commit their cells to particular fates
123. Nuclear control of cell morphogenesis and the interaction of nucleus and cytoplasm are well studied in Acetabularia.
124. Upon fertilization, the intracellular calcium ion concentration of the egg increases greatly
125. Bacteria are able to transmit genes from one individual to another by means of sex pili
126. +ve and – ve mating take place *Chlamydomonas*

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